EXPLORATORY DATA ANALYSIS USING POWER BI

WEEK 8 – STUDENT PERFORMANCE DASHBOARD REPORT

SUBMITTED BY:

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1. INTRODUCTION

The Student Performance Dashboard was developed using **Power BI** to analyse and visualize students' academic performance. The dashboard provides an interactive view of subject-wise scores, overall performance, parental education, test preparation, lunch type, gender, race/ethnicity, and performance grades.

The dashboard was created using the cleaned and transformed dataset prepared during Week 7. Different charts, KPI cards, slicers, and performance measures were used to make the analysis easy to understand.

2. OBJECTIVES

The main objectives of this activity are:

- To create an interactive Student Performance Dashboard using Power BI.
- To analyse the overall academic performance of students.
- To compare Math, Reading, and Writing performance.

- To analyse performance based on parental education.
- To compare students who completed test preparation with those who did not.
- To analyse performance based on lunch type.
- To examine performance across gender and race/ethnicity.
- To visualize the distribution of student performance grades.
- To use KPI cards and charts for quick analysis.
- To provide meaningful insights from the dashboard.

3. TOOLS USED

- **Power BI Desktop**
- **Power Query**
- **DAX**
- Student Performance Dataset

4. DATA VISUALIZATIONS:

4.1 KPI CARDS

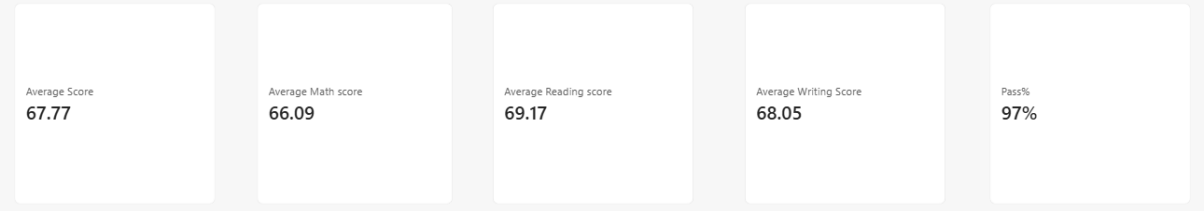
KPI cards were created to provide a quick summary of student performance.

The dashboard displays the following important values:

KPI	**Value**
Average Math Score	66.09
Average Reading Score	69.17
Average Writing Score	68.05
Overall Average Score	67.77
Pass Percentage	97%

These KPI cards provide an immediate overview of the academic performance of the students.

Screenshot - KPI Cards



4.1. RACE/ETHNICITY ANALYSIS

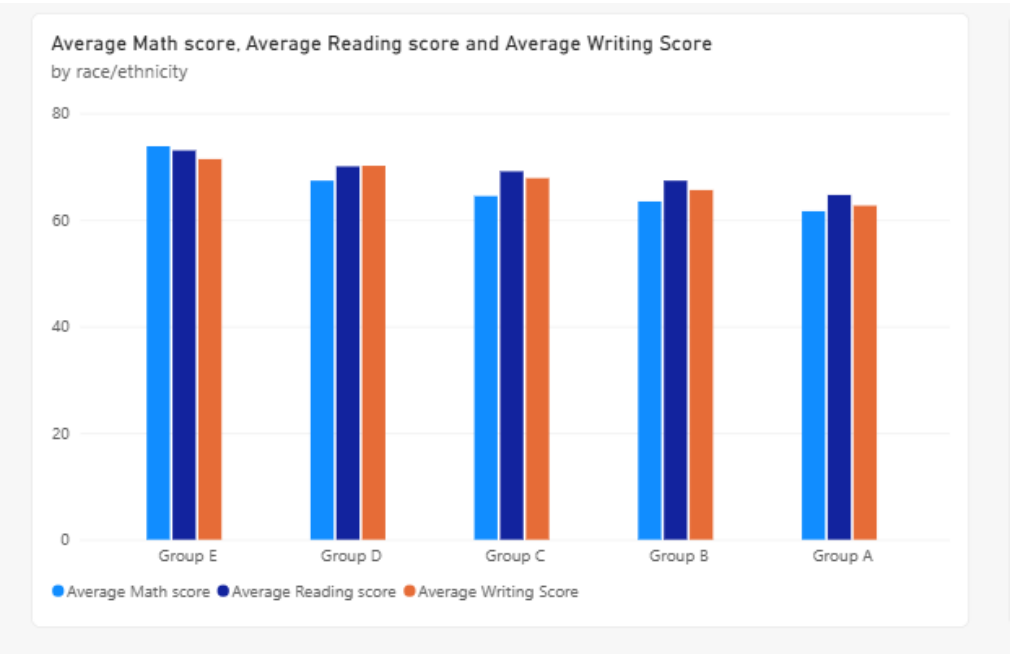
A Clustered column chart was included to analyse three Average scores by **Race/Ethnicity**.

The available categories include:

- Group A
- Group B
- Group C
- Group D
- Group E

This allows users to examine how the performance measures change when a particular group is selected.

Screenshot - Clustered column chart



4.3. PARENTAL EDUCATION ANALYSIS

A visual was created to analyse the relationship between parental level of education and students' average scores.

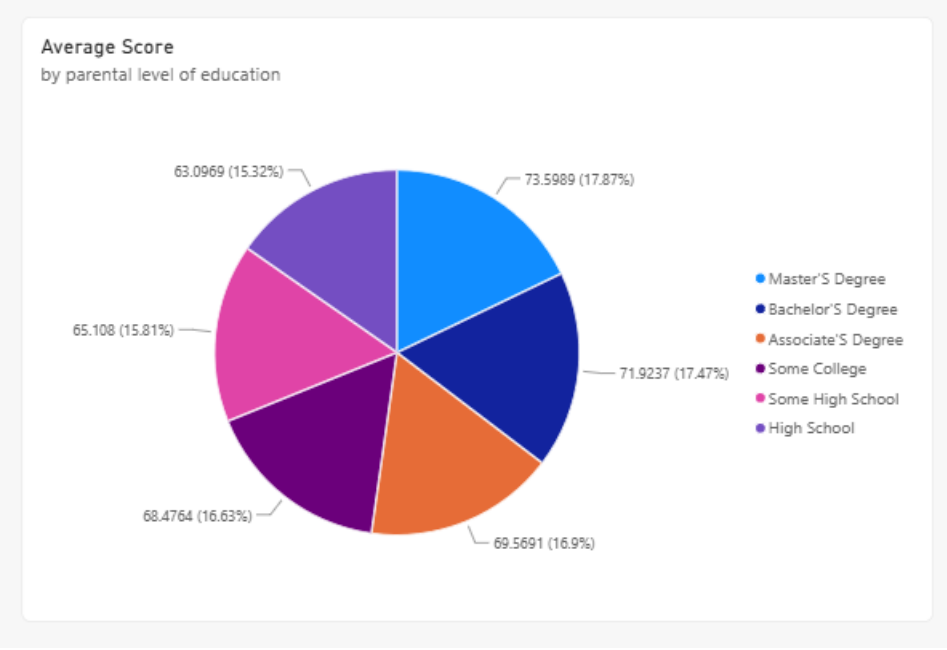
The dashboard compares different parental education categories such as:

- Associate's Degree
- Bachelor's Degree
- High School
- Master's Degree
- Some College
- Some High School

This visualization helps understand how average student performance varies across different parental education groups.

The dashboard shows differences in average scores between the groups, which can be used for comparative analysis.

Screenshot – Parental Education Analysis(pie chart)



4.4. SUBJECT PERFORMANCE ANALYSIS

A chart was created to compare the average Math, Reading, and Writing scores.

The dashboard shows:

- Average Math Score
- Average Reading Score
- Average Writing Score

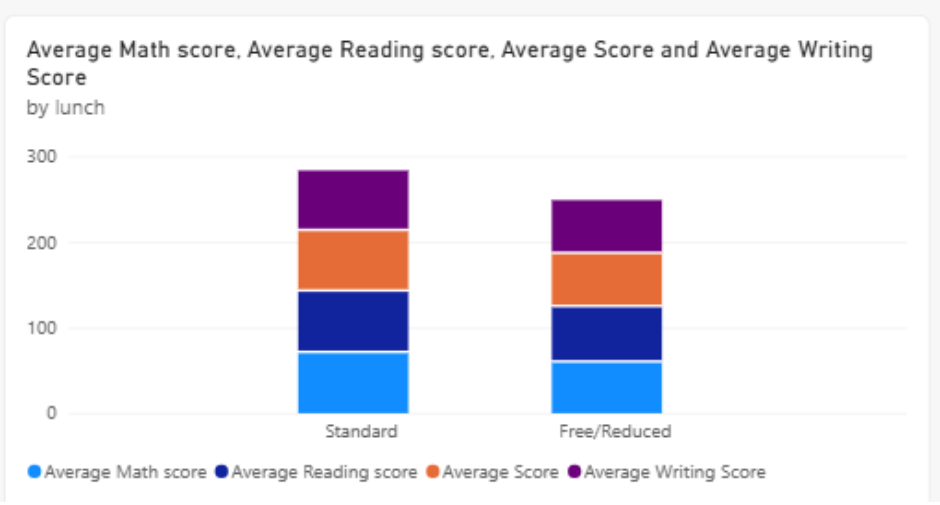
The subject comparison helps identify differences in student performance across the three subjects by lunch.

Based on the KPI values:

- Math Average = **66.09**
- Reading Average = **69.17**
- Writing Average = **68.05**

Reading has the highest average score among the three subjects.

Screenshot – Subject Performance(stacked column chart)



4.5. TEST PREPARATION ANALYSIS

A comparison was created between students who:

- Completed the test preparation course

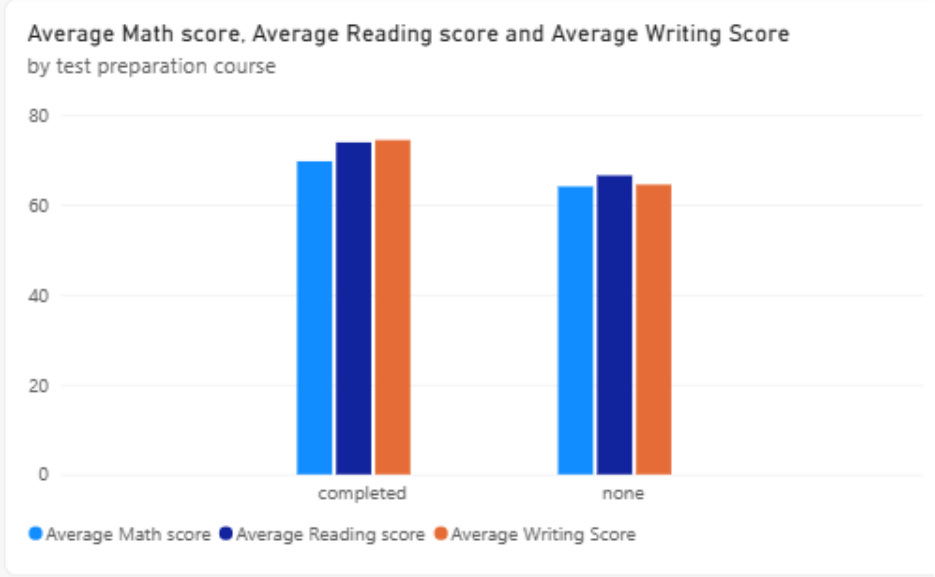
- Did not complete the test preparation course

The dashboard compares their Math, Reading, and Writing performance.

The visual helps determine whether the students who completed the preparation course had different average scores compared with students who did not complete it.

This analysis shows an **association** between test preparation status and performance; it does not by itself establish that the course caused the difference.

Screenshot – Test Preparation Analysis



4.6. GRADE DISTRIBUTION

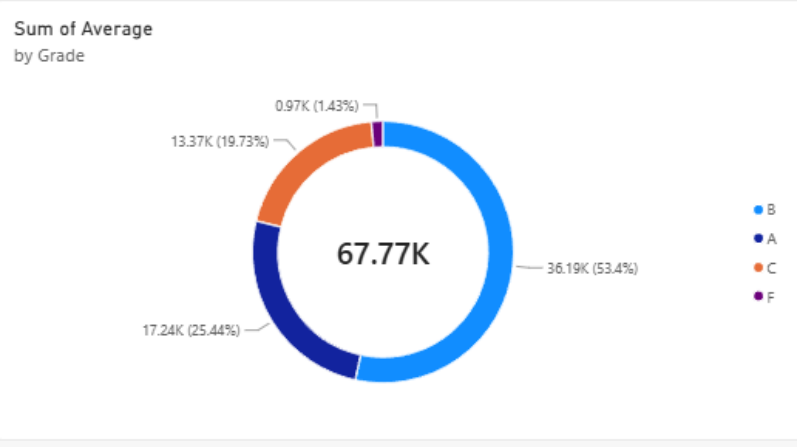
A Donut chart was used to display the distribution of students according to their performance grades.

The grades included are:

- A
- B
- C
- F

The visualization makes it easy to understand the proportion of students belonging to each performance category.

Screenshot – Grade Distribution



4.7. SLICERS

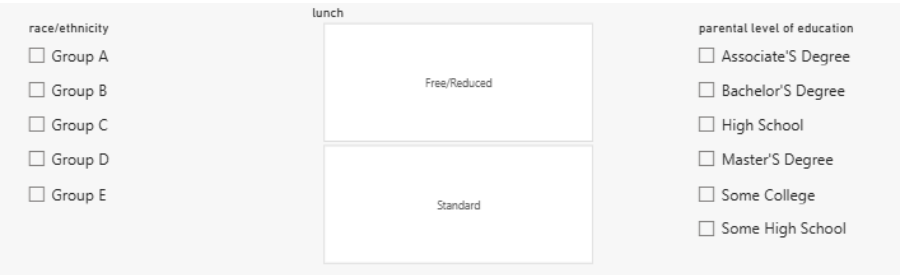
Interactive slicers were included to make the dashboard dynamic.

The dashboard contains filters for:

- Race/Ethnicity
- Lunch
- Parental Level of Education

When a slicer is selected, the related charts and KPI values update according to the selected category.

Screenshot – Dashboard Slicers



4.8. LUNCH TYPE ANALYSIS

The dashboard includes **Lunch** as a category for analyzing student performance.

The available lunch categories include:

- Free/Reduced

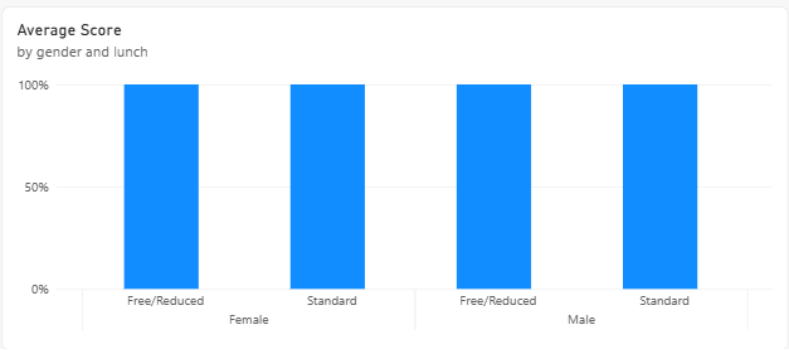
- Standard

The dashboard compares average scores based on lunch type.

This provides a way to examine differences in academic performance between the two lunch categories for each gender separately.

The visual should be interpreted as a comparison between groups rather than as evidence of a causal relationship.

Screenshot – Lunch Analysis



4.9. GENDER ANALYSIS

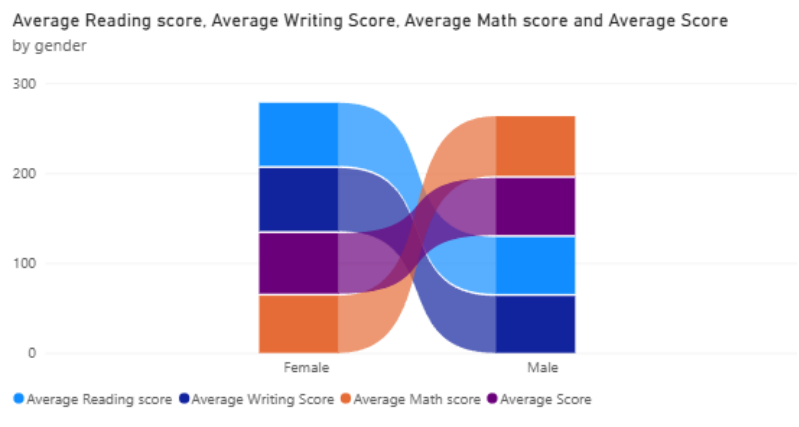
The dashboard also includes an analysis based on gender.

The visual compares average scores for:

- Female
- Male

The dashboard allows comparison of overall average performance between the two groups.

Screenshot – Gender Analysis



5. DASHBOARD VALIDATION

The dashboard was checked to ensure that:

- KPI cards display the correct values.
- Subject averages are calculated correctly.
- Pass percentage is displayed as a percentage.
- Charts contain appropriate fields.
- Slicers filter the dashboard correctly.
- There are no unwanted blank values.
- Visual titles are meaningful.
- The dashboard is easy to understand.
- Data values are consistent with the prepared dataset.

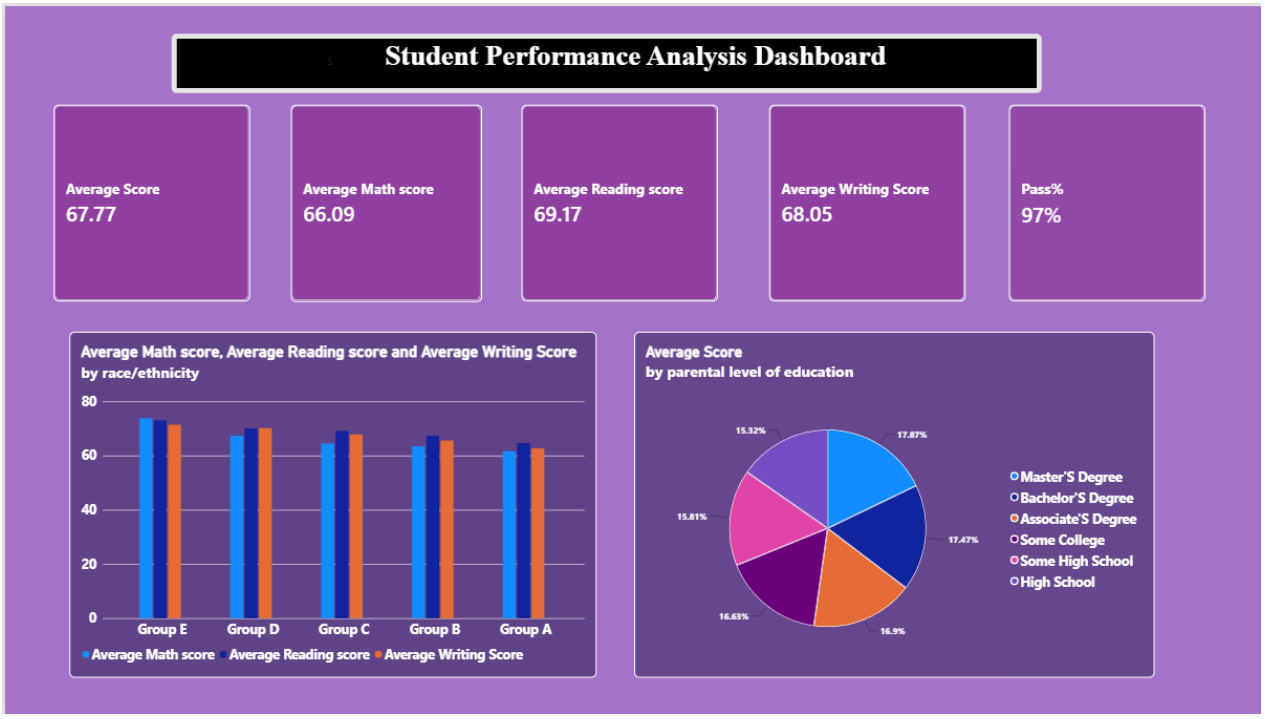
6. DASHBOARD DESIGN

The dashboard was designed using KPI cards, charts, slicers, and interactive visualizations.

The major sections of the dashboard include:

1. KPI Cards
2. Subject Performance Analysis
3. Parental Education Analysis
4. Test Preparation Analysis
5. Grade Distribution
6. Race/Ethnicity Analysis
7. Lunch Analysis
8. Gender Analysis
9. Interactive Slicers

Overall Student Performance Dashboard:







7. RESULT

The **Student Performance Dashboard** was successfully created using Power BI.

The dashboard provides an interactive view of:

- Overall academic performance
- Subject-wise performance
- Pass percentage
- Performance grades
- Parental education
- Test preparation
- Lunch category
- Gender
- Race/Ethnicity

The KPI cards and visualizations make it easier to understand student performance and compare different student groups.

8. CONCLUSION

The Week 8 activity provided practical experience in creating an interactive dashboard using Power BI. The cleaned dataset from Week 7 was transformed into meaningful visualizations using KPI cards, charts, slicers, and DAX measures.

The dashboard successfully summarizes student academic performance and allows users to explore performance across different demographic and educational factors. The analysis shows an overall average score of **67.77** and a pass percentage of **97%**, with Reading having the highest average among the three subjects shown.

Overall, the Student Performance Dashboard provides a simple and effective way to analyze academic data and identify important patterns in student performance.